

Healthcare Analytics Dashboard (Power BI)

1. Project Overview

A US-based healthcare organization wants a Power BI dashboard to monitor hospital performance, patient encounters, treatment costs, and insurance claims.

Currently, data is available in different systems but not centralized for reporting and decision-making.

The goal is to build a single dashboard to track KPIs and trends.

2. Business Objective

The dashboard should help management to:

- Monitor patient volume and hospital performance
 - Track treatment costs and revenue
 - Analyze insurance claim approvals/denials
 - Identify high readmission rates
 - Measure patient satisfaction
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3. Scope

In Scope

- Patient encounter analysis
 - Cost and financial tracking
 - Claims analysis
 - Satisfaction tracking
 - Time-based trend reporting
 - Drill-down analysis
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4. Data Sources

Data will be provided in CSV/Excel format:

- Patient Encounter Data (Fact Table)
- Patient Demographics
- Provider Details
- Hospital Information

- Date/Calendar Table
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5. Data Volume & Granularity

- ~200,000+ records
 - Data for multiple years
 - One row = one patient encounter
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6. Key Business Questions

- What is the total treatment cost trend?
 - Which hospitals have the highest readmission rates?
 - What % of claims are approved or denied?
 - Average length of stay by hospital?
 - Patient satisfaction by hospital and state?
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7. KPIs Required

Patient KPIs

- Total Patients
- Total Encounters
- Avg Length of Stay
- Readmission Rate %

Financial KPIs

- Total Treatment Cost
- Avg Cost per Encounter
- Insurance vs Patient Pay

Claims KPIs

- Claim Approval %
- Claim Denial %
- Pending Claims

Quality KPIs

- Avg Satisfaction Score
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8. Functional Requirements

Dashboard should include:

Filters/Slicers

- Date
- Hospital
- State
- Specialty
- Claim Status

Drill-downs

- Year → Quarter → Month
- State → City → Hospital

Pages

1. Executive Summary
 2. Hospital Performance
 3. Financial Analysis
 4. Patient Insights
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9. Data Preparation Requirements (Power Query)

Data must be cleaned for:

- Missing values
 - Duplicate records
 - Date format issues
 - Standardizing state names
 - Handling outliers in cost
 - Null satisfaction scores
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10. Non-Functional Requirements

- Dashboard load time < 5 seconds
- Simple and user-friendly UI
- Secure data (no personal info)
- Optimized data model

11. Deliverables

- Power BI Dataset
- Power BI Dashboard
- KPI List Documentation

12. Timeline (Sample)

Phase	Duration
Requirement Gathering	3–5 days
Data Cleaning	1 week
Modeling	3–4 days
Dashboard Build	1 week